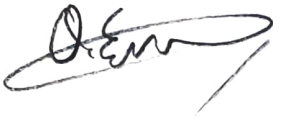
On my honor, I have neither given nor recheived any unauthorized andror inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of those principles will lead to a zero grade and is subject to harsh dicipline issues

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1)

$$\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 0 & (5-3\alpha)/(-12\gamma) & 9-2\alpha & \\ 0 & (\beta-2\frac{2}{3})(3+2\gamma)(\frac{\gamma}{3}) & -2\frac{8}{3} & \\ 0 & 14-\frac{8}{3}-6\gamma & 2-\frac{\gamma}{3} & \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & \frac{8}{3} & -\gamma & \frac{8}{3} \\ 0 & -\frac{49}{3} & -17\gamma & -21 \\ 0 & \beta-\frac{16}{3} & 3+2\gamma & \frac{16}{9} \\ 0 & \frac{34}{3} & -6+\gamma & -\frac{2}{3} \end{bmatrix}$$

$3\alpha-8=0$ $\quad 2+m=0$ $\quad 5-\frac{64}{3}$ $\quad -\frac{49}{3}$ $\quad \frac{64}{9}-\frac{16}{3}=\frac{16}{9}$

$3\alpha=8$ $\quad m=-2$ $\quad k=\frac{34}{43}$ $\quad 1-8.\frac{8}{3}$

$\alpha=\frac{8}{3}$ $\quad \beta=-11$

$\beta-\frac{16}{3}+\frac{49}{3}=0$

$\alpha=\frac{8}{3}$
$\beta=-11$
$\gamma=\frac{-215}{629}$
$m=-2$
$k=\frac{34}{43}$

$-17.\frac{-215}{629}$

$\frac{-215}{629}=\gamma$ $\quad \beta=\frac{-33}{3}-\frac{16}{3}+\frac{49}{3}=0$

$$\begin{bmatrix} 1 & \frac{8}{3} & \frac{-215}{629} & \frac{8}{3} \\ 0 & -\frac{49}{3} & \frac{3655}{629} & -21 \\ 0 & 0 & \frac{1455}{629} & \frac{205}{9} \\ 0 & 0 & 0 & -\frac{21667}{774} \end{bmatrix} = U$$

$3+2\gamma+17\gamma$ $\quad \frac{34}{3}-\frac{436}{3}=0$

$k=\frac{34}{43}.-17\gamma$

$3+2\gamma$ $\quad +\frac{-215}{629}.2$ $\quad -430$ $\quad \frac{1455}{629}$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & \frac{8}{3} & -\gamma & \frac{8}{3} \\ 0 & -\frac{49}{3} & -17\gamma & -21 \\ 0 & \beta-\frac{16}{3} & 3+2\gamma & \frac{16}{9} \\ 0 & \frac{34}{3} & -6+\gamma & -\frac{2}{3} \end{bmatrix} = \begin{bmatrix} 1 & \frac{8}{3} & -\gamma & \frac{8}{3} \\ 0 & -\frac{49}{3} & -17\gamma & -21 \\ 0 & 0 & 3+19\gamma & \frac{16}{9}+21 \\ 0 & 0 & & -21.\frac{34}{43} \end{bmatrix}$$

$-\frac{436}{3}k+\frac{34}{3}=0$ $\quad -17\gamma.k-6+\gamma$ $\quad -21k-\frac{2}{3}$ $\quad -\frac{205}{9.2}+\frac{-714}{43}$ $\quad \frac{-21667}{229}$

$\frac{34}{43}.-17.\gamma-6+\gamma$ $\quad -21.\frac{34}{43}$ $\quad \frac{-8815-12852}{43.18}$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1/2 & 1 \end{bmatrix} \times \quad -\frac{3+19\gamma}{2}+\frac{34}{43}.-17.\gamma-6+\gamma=0$$

$317$ $\quad 1156$

$-\frac{3}{2}-\frac{19\gamma}{2}-\frac{578\gamma}{43}-6+\gamma=0$

$(43)$ $\quad (21$

$-\frac{15}{2}-\frac{1973\gamma}{86}+\gamma$

$\frac{-215}{629}=\gamma$ $\quad \frac{629}{43}\gamma=-5.\frac{43}{629}$ $\quad \frac{1887}{86}\gamma=-\frac{15}{2}$ $\quad -\frac{15}{2}-\frac{1887\gamma}{86}=0$

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